



A systematic literature review on explainable AI for churn prediction, customer segmentation and retention

Misbah Maqbool¹ · Ishtiaq Hussain Qureshi¹

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Abstract

Nowadays, the incorporation of XAI—explainable artificial intelligence—into customer analytics has gained major attraction driven by the need for trust, transparency and actionable understanding in predictive modelling. This systematic literature review analyses the application of XAI techniques like SHAP, LIME and also other AI models like machine learning, deep learning, ensemble models and further provides counterfactual explanations in churn prediction, customer segmentation and retention strategies. Using a structured search across important databases like Scopus, IEEE and other digital libraries, 80 studies published between 2015 and 2025 were selected and analysed. This review categorises XAI approaches by industry application, interpretability findings and methodology and emphasises their role in improving model decision-making and transparency. The findings explore that XAI not only enhances stakeholder trust but also enables more ethical and targeted customer interventions. A comparative study has also been performed for XAI techniques for churn prediction, customer segmentation and retention. The limitations in real-time deployment, standardisation and evaluation metrics are discussed in gaps and discussions along with future research directions for XAI models, hybrid models and domain-specific explainability contexts.

Keywords XAI · Churn prediction · Customer segmentation · Customer retention · Machine learning models · Deep learning models · Ensemble models

1 Introduction

1.1 Purpose of the study

Customer churn prediction is important in recent CRM—customer relationship management—supporting organisations, which proactively identify at-risk customers and increase the lifetime value of customers. Important revenue losses resulted from greater churn rates, which happen in subscription-based telecommunications and services, education, healthcare, retail, insurance, banking and other organisations that are highly based on data-driven methods to improve customer retention [22–24, 49]. While businesses collect a broader amount of customer data, they collect

actionable insights from these challenging datasets. Data mining is a major restraint in AI, enabling enterprises to uncover trends and hidden patterns in the behaviour of churn. But the churn prediction models' effectiveness differentiates mainly based on methodology options, characteristics of datasets and industry-specific factors. This review thus reviews more research articles published between 2020 and 2025, concentrating on explainable AI for churn prediction, customer segmentation and retention. The review is as follows,

- Investigate various churn prediction methods across different industries.
- Examines explainable AI, ML and DL techniques as comparative performance in churn prediction, exploring the XAI approaches like LIME, SHAP and counterfactuals used for churn predictions.
- Assesses general challenges like data imbalance, interpretability, feature selection and drifts in concepts
- Emphasises emerging trends related to explainable AI in churn prediction and customer retention and discusses the

✉ Misbah Maqbool
misbah.msscholar@kashmiruniversity.net

Ishtiaq Hussain Qureshi
ishtiaq@yaho.co.in

¹ Department of Management Studies, University of Kashmir, Srinagar 190006, India

transparency and effectiveness of these models in real-world applications.

- Explores gaps and future recommendations from the review.

Churn prediction is important in establishing effective segmentation and retention approaches, allowing businesses to get ahead of customer attrition, allocate retention budgets and personalise marketing efforts effectively. Research recommends that companies executing advanced churn prediction approaches can enhance retention rates by 5–10 percent, resulting in profit increases of 25–95 percent [3]. [23] focused on comparative study initially using data sampling approaches and hyperparameter optimisation methods in ML models for churn prediction, emphasising their influence on predictive performance. Further, the analysis extended to analysing the efficacy of ADASYN, GNUS and SMOTE upsampling methods in combination with XGBoost and random forest machine learning algorithms under various class imbalance levels. [19] provides a broad survey of ML algorithms for churn prediction, arguing their limitations, practical applications and strengths. [17] concentrates on hyperparameter tuning on deep learning based churn prediction models, specifically in banking, offering insights into optimising deep neural networks (DNNs) for enhanced accuracy. These studies provide valuable views on churn prediction study's challenges and methodological advancements.

The major purpose of this research is to review and combine existing studies on the XAI—explainable AI—application methods in churn prediction, customer segmentation and retention strategies. The conventional ML models established greater predictive accuracy, but the interpretability is limited, which is critical for decision makers in customer-centric organisations [39]. In connecting the gap between interpretability and predictive performance, this research plans to authorise businesses with actionable understandings which are both understandable and accurate, thereby improving compliance, customer-centric decision-making and trust [6, 25, 34]. In comparison with [19], which focuses on general churn prediction analysis using ensemble machine learning methods and hence this study focuses on recommendation of these existing studies as that advocate the use of ensembles and deep learning techniques, and as well as the adoption of explainable methods to drive further advancements, considered as a novelty of this review. In combining the understandings from recent studies, this review serves as a valuable source for researchers, industry practitioners and data scientists, supporting them in understanding better practices, future directions and methodological advancements in churn prediction.

1.2 Need for churn prediction in business

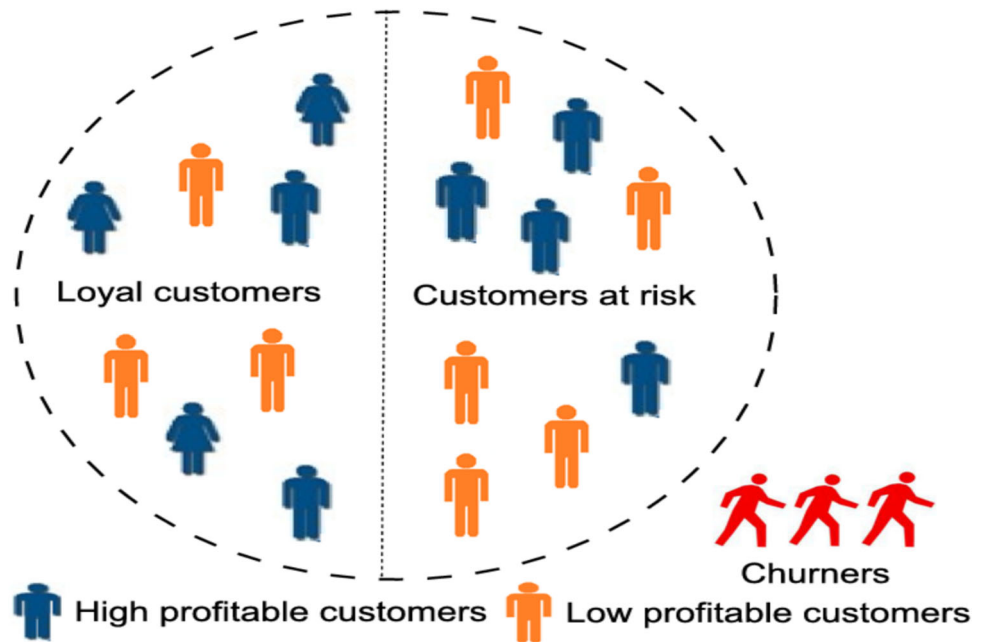
In a highly competitive global market, a business's economic growth is based on increasing consumption and extending the existing consumers' average lifespan. In attaining customer retention, it is important to understand the factors contributing to predicting customers' churning intentions and customer loyalty development, and for that, effective retention campaigns performed are shown in Fig. 1.

In understanding this process, businesses can establish the strategies to hold the existing customers and establish stronger loyalty bonds with them, which is less costly and highly effective compared with attracting new customers. Losing profitable customers can be expensive, resulting in a transformation in market pattern from the importance of attaining newer customers to holding existing ones. Moreover, churn impact circulates in social infrastructure as customer mutual interdependence exists, and when churn happens, it raises churn probability in the social environment [33]. In spite of the efforts of the organisation to establish customer loyalty, it faces challenges because of the exposure of customers to challenging market awareness and promoted offers. In assuring the growth and sustainability of business, it is important to understand the framework of establishing customer loyalty bonds [18]. Customers are the major assets of any organisation, and businesses are responsible for explaining and executing policies which extend their lifetime. Customer lifetimes extension must be a major focus of companies, and it governs their future commercial direction.

Forcing customers into purchasing upgrades without satisfying their real requirements resulted in greater customer rejection and reduced customer satisfaction, eventually damaging business status [3]. Moreover, the end of unprofitable customers can have an unplanned negative impact on the loyalty of retained valuable customers [57]. Restricting customers' freedom to switch through contractual restrictions is said to be an opposite method in handling the relationship of loyalty satisfaction [10, 43]. But these restrictions can negatively influence customer loyalty and are not suitable for the long term. Further, it is based on the customer's intention, and due to the heterogeneous nature of customer intention and service requirements, it is impossible to build a loyalty connection with all customers.

Customers do not churn generally because of individual dissatisfaction with the setup, and there are usually several examples of dissatisfaction before the customer ends the transaction with the business. Subsequently, dissatisfied customers typically present, and so organisations concentrate on establishing retention campaigns in enhancing their offers and executing better switching barriers. Figure 1 depicts the customer churn lifecycle, dividing customers into loyal at risk and churn segmentation. It highlights the significance of

Fig. 1 Customer churning intentions and customer loyalty development



stratifying customers with respect to profitability, discriminating among highly and non-highly profitable segments. The major goal of customer churn prediction is to alleviate churn, especially in highly profitable customers, guiding retention efforts towards those that importantly influence profitability. Significantly, businesses must adapt their strategies, and getting feedback from all customers can be costly and usually tend to analyse representative samples. When establishing retention approaches, companies must focus on the expected lifetime value of every customer and relate it to the costs of the retention efforts. Unprofitable customers retaining who do not defend the retention campaign cost may not be considered cost-effective for the business. Hence, it is important in balancing the expenses and benefits of retention campaigns to make informed decisions. It is highly advantageous for businesses to concentrate their resources on holding a targeted group of customers compared to trying to increase total retention rates of all customers.

Customer churn is said to be the occurrence of customers discontinuing their relationship with a business, which is a critical challenge across sectors like banking, telecom, SaaS and e-commerce. The cost of getting a new customer is estimated to be 5–7 times greater than retaining the existing customer. Hence, predicting churn precisely and earlier is important for maintaining gains and customer loyalty.

From the studies, the growing need for churn prediction includes,

- Targeted retention campaigns: Predictive models help identify high-risk customers, enabling personalised interventions.

- Competitive advantage: In saturated markets, proactive churn management differentiates leaders from laggards.
- Revenue protection: Churn directly impacts recurring revenue streams. Even a 5% increase in retention can boost profits by 25–95% [39].
- Customer lifetime value (CLV) optimisation: Retaining high-value customers enhances long-term business growth.

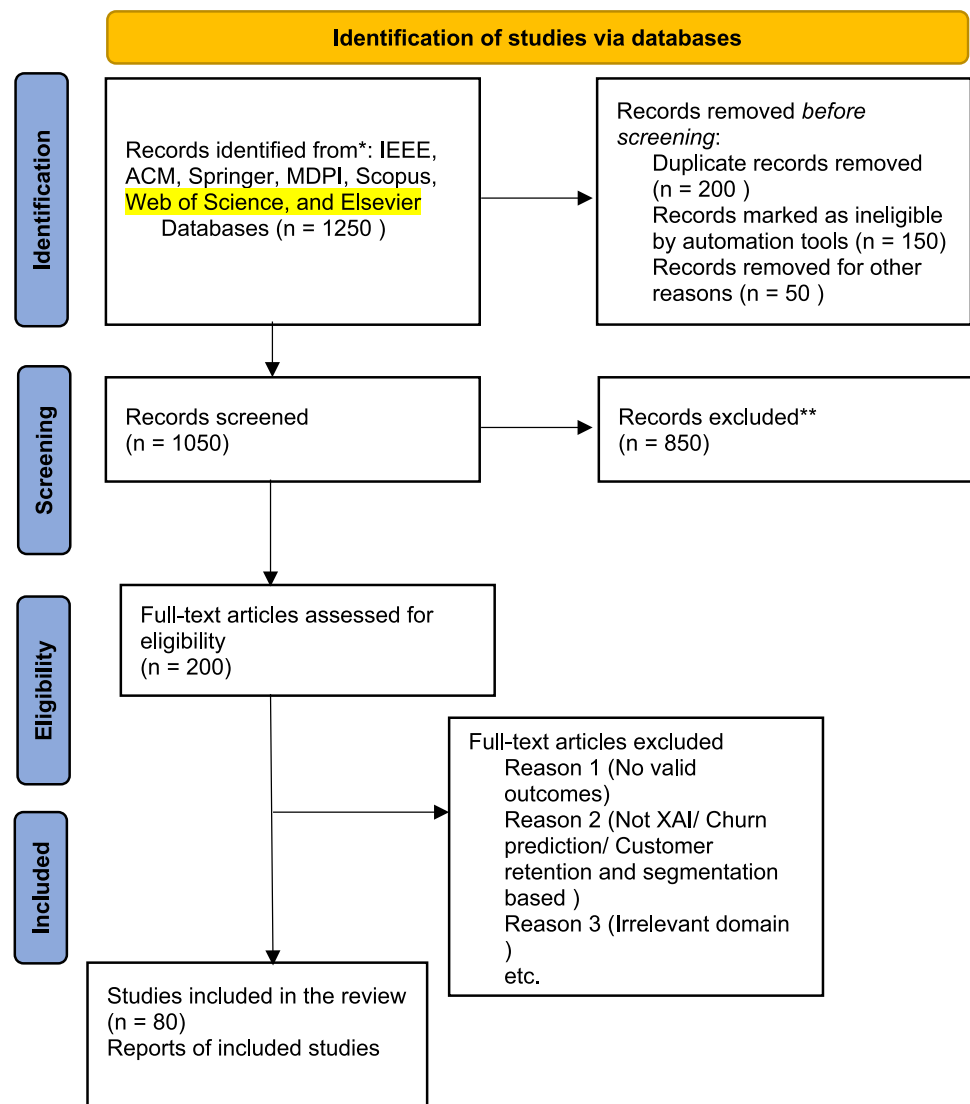
But the existing churn models usually lack transparency, making it difficult for stakeholders to understand why the customer is likely to churn. Hence, XAI becomes vital, providing interpretable insights which are helpful in regulatory-compliant, customer-aligned and ethical approaches [6, 25, 34].

2 Methodology

The methodology focuses on collecting articles with search strategies and creating a PRISMA for performing a literature review.

2.1 Articles collection/search strategies

The main core of the systematic literature review is article collection or search strategies, and thus, this review will be comprehensive and free from bias. The clear step-by-step breakdown of an effective search strategy is framed in Table 1.

Fig. 2 PRISMA outline (Source: Author)

The structured approach collects and filters the journals or articles based on explainable AI, AI in churn prediction, customer retention and segmentation strategies. Using Boolean logic, it combines all articles and focuses on the search string. The searching continues across major academic databases like IEEE, Scopus, Web of Science, Springer, MDPI and others, applying filters for the duration 2015–2025, and the language should be in English. At the same time, documenting the number of records, inclusion and exclusion criteria is mentioned in the PRISMA diagram, also shown in Fig. 2.

2.2 Flow chart/PRISMA outline

3 Predictive model evaluation for churn prediction, customer segmentation and retention

Churn prediction uses machine learning (like fuzzy clustering, logistic regression, random forest, XG Boost, xAI through SHAP and Lime) to identify customers likely to leave. Evaluation metrics include accuracy, precision, recall, F1-score and ROC-AUC. Customer segmentation uses clustering techniques (e.g. *K*-means, hierarchical) to group customers based on behaviour, demographics or value, enabling personalised strategies. Retention strategy evaluation uses interventions (like offers or campaigns) that are tested using A/B testing, lift analysis and CLV to measure effectiveness in reducing churn and increasing loyalty. Integrated frameworks combine prediction, segmentation

Table 1 Articles collection/search strategies step by step

Step	Details
Research objective	To identify and organise the studies related to Explainable AI, AI in churn prediction, customer retention and segmentation strategies
Key concepts	• XAI or explainable AI • Churn prediction • Customer segmentation • Customer retention
Synonyms/Keywords	• “Explainable AI”, “XAI”, “model interpretability”, “Interpretable models” • “AI churn prediction”, “customer segmentation”, “customer attrition”, “user clustering”, • “retention analysis”, “loyalty prediction”
Boolean search string	(“Explainable AI” or “XAI” or “interpretable models”) AND (“churn prediction” OR “customer segmentation” OR “customer retention”)
Databases searched	IEEE, ACM, Springer, Scopus, Web of Science, MDPI, Elsevier
- Search filters applied	Publication considered: 2015- 2025 Language: English Document type: Journal articles, Conference papers
Tools used	Endnote or Excel for managing references
Inclusion criteria	Studies involving XAI, customer-related predictive models, focus mainly on AI churn prediction, segmentation, and retention. (n = 1050)
Exclusion criteria	Studies not using XAI, AI, for churn prediction, in non-English journals (n = 850)
Screening process	Title and abstract Full text eligibility review Documentation in PRISMA
Documentation	Numbers recorded for every stage Identified, screened, excluded, included Justification for exclusion is clearly noted

Source: Author

and personalised recommendations to boost customer satisfaction and long-term profitability. The following section discusses the predictive model evaluation of customer segmentation through fuzzy clustering, churn prediction through XGBoost model and explainability of churn through SHAP and Lime [27, 29, 36, 55].

3.1 Machine learning techniques (existing techniques like fuzzy clustering, XGBoost)

The progressing difficulties and massive customer data have demanded a paradigm transformation from traditional statistical approaches to highly flexible and adaptive machine learning (ML) methods in customer retention analytics. Conventional methods like discriminant analysis and logistic regression are helpful for interpretable and linear relations, lacking in the capability to model the nonlinear interactions or identify higher-dimensional and complex patterns. As datasets became better with transactional, unstructured and behavioural text data, ML methods began to outperform traditional methods in both feature discovery and predictive accuracy [16]. Customer-centric data arise from digital platforms, which encouraged the transformation in which the retention methods expected personalised and real-time understandings across several targets [38].

ML enables automated learning from the existing datasets, making it specifically helpful in dynamic customer landscapes like banking, e-commerce and telecom. Methods like random forests, fuzzy clustering, gradient boosting and support vector machines (SVM) have shown greater robustness and adaptability in modelling churn prediction. Unlike rule-based models, which are based on predefined beginnings, ML methods are constantly adjusting to newer patterns, focusing on the drift in concept and difficult customer behaviours. In addition, advancements in data infrastructure and computing power, including GPU-accelerated training and cloud-based analytics, enabled the larger-scale deployments of ML-based retention models in both strategic and operational decision-making mechanisms. This transformation shows a major evolution in analytical capabilities available to industries focusing on reducing churn and encouraging longer-term customer engagement. Supervised learning methods remain a better method in ML-based churn prediction, in which the historical labelled data comprising of retained and churned customers are used to train models which simplify to unseen instances. Decision trees and their ensemble alternatives like gradient boosting machines and random forests have continuously demonstrated greater predictive performance in recognising the churn-prone consumers across different areas like finance, telecom and retail [28].

Decision trees split data recursively based on important features, generating interpretable decision paths, while random forests combine several trees to decrease variance and

enhance generalisability [47]. Gradient boosting, like XG boosting and other boost algorithms, improves the predictive power by orderly correcting the errors from earlier iterations, making the model for catching understated churn indicators. SVMs have been practical for higher-dimensional datasets using the kernel functions to classify nonlinear patterns in the behaviour of customers. LASSO or ridge—logistic regression—is still used as the state of the art in machine learning research, specifically when interpretability is required. Neural networks like MLPs resulted in better performance when the complexity of the data requires hierarchical, deep learning structures. These trained supervised algorithms use evaluation metrics like recall, AUC-ROC and accuracy for assuring robustness across challenges in class imbalance. The researchers have established that ML methods continuously perform better than conventional statistical methods in predictive accuracy, specifically when cross-validation approaches and engineered features are applied effectively. The flexibility and scalability of these supervised methods have allowed firms to execute churn models in real-time decision systems, which drive personalised retention approaches at scale.

Machine learning incorporation into sector-level retention systems, which recent technologies in explainability, model automation and deployment mechanisms have enabled. Sectors are recently used end-to-end ML pipelines which automate data pre-processing, feature selection, training of models and prediction scoring within the CRM models [30]. Hyperparameter tuning and model selection streamlined by AutoML tools facilitated non-expert users to arrange highly accurate models with minimal manual intervention. These tools usually integrate ensemble methods and performance validation procedures, decreasing the poor generalisation and overfitting risks. Cloud-based platforms like Microsoft Azure ML, AWS SageMaker, Cloud AI have reduced the technical errors to users retention models at scale, offering potential settings for monitoring, model training and inference.

Explainability tools like LIME—local interpretable model-agnostic explanations—and SHAP—shapley additive explanations—have dealt with “black-box” focus conventionally related to recent ML models, providing transparency into decision-making processes and sustaining regulatory compliance standards. Model deployment in churn analytics has been fruitful across different industries. Predictive ML models are fixed into the customer service models for prompting real-time retention proposals in telecommunications. ML-based personalisation engines, e-commerce platforms enthusiastically engage high-risk users again and suggest products. Churn scoring models and real-time credit risk models are incorporated with call centres and mobile apps to enhance customer service outcomes [13]. The cloud infrastructure, business intelligence systems and ML convergence have facilitated continuous learning, operationalised

retention analytics, predictive understandings of unified scaling and campaign optimisation within workflows of firms [35].

3.2 Deep learning techniques

Deep learning application to churn prediction started as sectors combined high-granularity clickstream, usage and sentiment data that exceeded the representational capacity of shallow models. MLP rapidly gave way to deep recurrent and convolutional architectures capable of pulling out hierarchical thoughts from raw behavioural records [46]. CNNs were initially promoted for image recognition to obtain local activation patterns in orderly service usage matrices, facilitating telcos to identify understated frequency transformation of previous defection. Autoencoders use representation learning by compacting higher-dimensional transaction vectors into solid latent codes that save monetary value and buy periodicity while performing noise filtering.

The researchers comparing CNN-based churn detectors with gradient boosting machines stated gains in AUC-ROC when there is available unstructured log data, highlighting the benefits of end-to-end feature extraction. Batch normalisation and dropout regularisation accelerate convergence and mitigate overfitting, making deep models feasible on organisational hardware. In addition, residual connections promoted by ResNet variants permitted in-depth stacks without the issue of vanishing gradients, enabling richer temporal abstractions throughout the customer expeditions [1]. Mutually, these improvements enabled experts to consume loyalty scores, raw event streams and other complaints into a combined pipeline receiving granular churn probabilities at scale. The studies continuously demonstrate that when data variability and volume are higher, deep neural frameworks exceed traditional and trivial learners by identifying nonlinear signals like complaint tone transformation or session volatility, which pave the way into observable detachment. Since churn is integrally a temporal sensation, RNNs—recurrent neural networks—and their gated extensions have attained importance in sequential customer behaviour modelling.

GRU—gated recurrent Unit—and LSTM—long short-term Memory—are designed to obtain a longer range of dependencies without the problem of vanishing gradients, which have been applied effectively to telecom records of call details and media logs subscriptions, attaining better recall of earlier stage churners related with fixed windows classifiers [32]. TCNs—temporal convolutional networks—improved sequential modelling through obtaining parallelisation benefits, in-depth receptive fields and replacing recurrence with dilated convolutions [48]. According to [45], attention mechanisms primarily proposed for the translation of neural

machine learning models have been adapted to weigh prominent events like billing anomalies or service outages and hence enhancing interpretability and encouraging F1 scores in churn detection. Moreover, hybrid models which combine LSTM backbones with attention layers validate notable gains in precision when predicting the involuntary versus voluntary attrition, in which the attention carries important contextual activities like competitor promotions taken in social feeds.

Hierarchical recurrent networks further disaggregate sequences into customer lifecycle and session level tiers, obtaining dissatisfaction micro bursts combined within macro usage trends. Comparative researches across e-commerce and banking datasets disclose that sequence-aware mechanisms can decrease false-negative churn predictions by till 30% compared to static feature models. These outcomes highlight that the modelling of temporal context value clearly permits forms to intervene during critical windows when actions of retention are highly effective. Advanced churn mechanisms are increasingly based on cultured representation learning methods that extend beyond feed-forward methods. Variational autoencoders have been leveraged to make smooth latent manifolds of customer behaviour, enabling early churn warnings through anomaly detection [45].

Embedding layers resulting from doc2vec and word2vec convert categorical features like product categories or tariff plans into dense vectors, which encode relational similarity and improve downstream classification performance [56]. GNNs—graph neural networks—model the interactions of customers and social impacts by treating users as communication and nodes combined as edges, and the empirical results show that integrating network centrality measures enhances churn AUC by obtaining peer impurity [4]. Promising capsule networks have been identified to preserve hierarchical methods for the entire relationships in multichannel behavioural data, obtaining favourable earlier findings in subscription media churn responsibilities [28]. Moreover, LIME and explainability tools like SHAP [56], predictive power have become vital to deep churn channels, offering customer-level rationales which support agent scripting and satisfy regulatory scrutiny. Layer-wise significance propagation views contribution scores across channels and time steps, helping managers to trace churn signals to differentiate events like negative reviews or declined payments [1]. Ensemble distillation methods compress complex frameworks into interpretable substitutes without major accuracy loss, linking the gap between managerial usability and advanced modelling. These developments collectively validate that interpretability mechanisms and cutting-edge representation learning can co-exist, obtaining transparent and high-yielding churn predictions [6, 35].

Table 2 Comparative table: XAI techniques in customer analytics [11, 40, 42]

Aspect	Customer segmentation & retention	Churn prediction
Common XAI techniques	SHAP, LIME, DECISION TREES, PARTIAL DEPENDENCE PLOTS	SHAP, LIME, interpreted ML partial dependence, counterfactual explanations
Model types used	K-means, DBSCAN, hierarchical clustering + XAI overlays	XGBoost, random forest, neural networks with post-hoc XAI
Interpretability focus	Segment profiling, feature importance for retention triggers	Feature attribution for churn risk, instance-level explanations
Stakeholder usability	Marketing teams use XAI to tailor retention campaigns	Compliance teams and analysts use XAI to justify churn predictions
Recent findings	SHAP helps identify high-value segments based on behavioural traits	SHAP outperforms LIME in consistency and stakeholder trust in telecom churn prediction
Fairness & bias detection	Limited studies; mostly focused on demographic fairness in segmentation	SHAP reveals bias in contract types and tenure influencing churn predictions
Prescriptive capabilities	Emerging use of XAI with optimisation to suggest retention actions	SHAP + optimisation used to recommend personalised retention strategies
Dataset types	Synthetic and real-world CRM datasets (e.g. e-commerce, banking)	Telecom, banking, and subscription datasets (e.g. 7,043 telecom records)

3.3 Comparative table (customer segmentation and retention and churn prediction using xAI techniques)

From the (Tables 2 and 3), it has been recognised that,
Churn prediction:

- Widely adopted SHAP and LIME techniques to reveal reasons behind customer churn.

Table 3 Comparative table on XAI techniques used, model type and description

Study & reference	Focus area	XAI techniques used	model type	Key insights
[40]	Churn prediction	SHAP, LIME	Random forest, logistic regression	SHAP provided more consistent and interpretable churn insights for compliance
[41]	Churn mitigation & retention	SHAP, LIME, counterfactual explanations (DiCEML)	Balanced random forest	Counterfactuals enabled actionable retention strategies and bias detection
[44]	Customer retention	Feature importance, decision trees	AdaBoost, Random Forest, MLP	Ensemble models with XAI improved retention targeting and campaign design
[8]	Churn prediction	Feature attribution, boosting	SVM, ANN, decision trees	Boosted SVM achieved the highest accuracy; XAI helped interpret churn drivers
[31]	Churn prediction & retention	Feature importance, deep RL	Deep RL, random forest	RL-based models offered dynamic retention strategies with interpretable outputs

Source: Author

- Models range from random forest to deep reinforcement learning, emphasising interpretable and actionable insights.
- Studies show SHAP provides consistent interpretations, aiding compliance and bias detection.
- Counterfactual explanations help formulate personalised strategies to prevent churn.

Customer segmentation & retention:

- XAI is used for profiling high-value customer segments and guiding retention campaigns [20].
- Techniques like feature importance and decision trees help marketers target the right audience.
- Ensemble and RL-based models enhance retention strategies with explainable and dynamic recommendations.

Therefore, SHAP emerges as the most trusted XAI method across both domains. Integrating XAI leads to more strategic and transparent decision-making. Some studies move beyond prediction to prescribe retention strategies using interpretable outputs. Explainable AI (XAI) methods such as SHAP, LIME, counterfactuals, rule lists and GAMs are increasingly applied to churn prediction, customer segmentation and retention. Their value lies not only in interpretability but also in robustness, reproducibility and managerial actionability. A “safe machine learning” perspective emphasises stress-testing explanations under noise, feature correlation

and distribution shifts, while ensuring feasibility and causal validity. SHAP and GAMs offer stable global insights, LIME provides intuitive but variable local stories, and counterfactuals guide actionable retention strategies. Industry applications from telecom to banking show XAI enhances trust, compliance and targeted interventions when robustness is systematically evaluated [15].

4 Gap analysis and recommendations

The present review explores the better performance of ML, deep learning, XAI approaches, ensemble models, specifically gradient boosting and fuzzy clustering, XAI in predicting customer churn across various industries and datasets. This finding aligns with previous research highlighting the ensemble methods’ robustness in managing higher-dimensional and complex data [10]. In contrast to conventional classifiers like decision trees, logistic regression and ensemble models, continuously demonstrated greater discriminatory power as shown by their elevated pooled F1 and AUC-ROC scores. These findings are similar to the conclusions of [12] and [50], who identified that ensemble methods obtain better predictive accuracy by capturing the nonlinear feature interactions automatically without the extensive manual feature engineering. Moreover, associatively lower variance in performance across research recommends that ensemble models provide constant predictive

capabilities, regardless of dataset heterogeneity. Compared with prior studies, which usually concentrated on limited datasets or specific sectors, this systematic literature review offers stronger cross-major generalisability, strengthening ensemble models, specifically gradient boosting and fuzzy clustering, xAI in predicting customer churn across various industries and datasets, as the preferred one for customer retention applications. But the support on tree-based ensembles also advances interpretability focus, resounding assessments by [14] who caution against the indiscriminating adoption of opaque models in spite of their predictive strength. Hence, future studies may explore hybrid methods that combine ensemble accuracy with explainability. Deep learning models like long short-term memory (LSTM), convolutional neural networks (CNNs) and recurrent neural networks (RNNs) emerged as the most relevant algorithms for datasets involving behavioural and time series sequences. These findings are consistent with [17], who identified that deep frameworks excel in extracting temporal dependencies from sequential customer data like clickstreams and app usage logs.

Previous studies like [2] emphasised that LSTM and RNN outperform traditional classifiers in digital platform and subscription-based landscapes because of their ability to model long-range dependencies. Preferably, deep learning models in the present meta-analysis attained the greatest F1 scores within behavioural datasets, pointing out their appropriateness for churn scenarios which demand accurate identification of detachment risks. The minimal need in manual feature engineering in deep learning mechanisms as observed by [59] and [14] validates their realism in large scale implementations. But it is worth pointing out that these models need important computational resources and address challenges in the deployment of real-time, a drawback also recognised by [13]. In spite of these challenges, the results uphold that deep learning models hold a central position in preferred areas characterised by dynamic customer behaviour, specifically when paired with recent environments like cloud-based and GPU analytics platforms. The major role of data pre-processing and feature engineering in improving model performance is another noticeable finding of this systematic literature review. Previous research [59] and [17] have highlighted the significance of crafting informative features like (RFM) recency, frequency and monetary metrics and contract details and compliant frequencies for better churn prediction. The findings verify the earlier understandings showing that models integrating the lag features, temporal aggregations and engagement deltas continuously perform better than those based on minimal processed inputs.

In addition [14] findings use class balancing methods like SMOTE, which importantly recall metrics in imbalanced churn datasets. Significantly, the [14] emphasised that data pre-processing steps, including encoding, normalisation and

outlier removal, contributed to important gains in F1 scores and AUC-ROC, specifically in ensemble and deep learning models. These findings strengthen the vision expressed by [5] that no algorithm can overcome poor quality of data, establishing thoughtful pre-processing as a vital predictive modelling component in channels. In comparison with earlier research, this literature review offers a more comprehensive quantification of pre-processing benefits across industries and algorithms, reinforcing the case for incorporating data engineering strategies in churn analytics [51]. From the findings of [14], it has been identified that the telecommunication sector mostly uses ensemble models and decision trees because of their operational efficiency and interpretability. Financial institutions and banking focus on interpretable models like EBM—explainable boosting machines—and logistic regressions, used by [52], replicating the regulatory compliance requirements. Alternatively, deep learning models rule the SaaS and e-commerce sectors in which high-frequency behavioural data and complex user interactions are common, similar to trends reported from [54].

Retail landscapes continue in uplift modelling and segment-based methods focusing on market-oriented and RFM metrics evaluations in accordance with [37]. These outcomes highlighted the importance of contextualising model selection with respect to customer behaviour patterns, operational goals and industry-specific dynamics. Dissimilar to previous research, which usually focused on individual firms, these findings provide a wider view, emphasising trade-offs among interpretability, deployment feasibility and accuracy in churn analytics. The findings of the research highlighted the impact of evaluation metric selection on the interpretation of model performance, as muffled by evaluations by [51] and [17] concerning the drawbacks of imbalanced datasets' accuracy. In line with previous studies [13], the broader adoption of LIME and SHAP tools in the reviewed research pointed out the growing demand for explainable AI in churn modelling. The evaluation shows that interpretability tools increase managerial trust and enable greater deployment rates, confirming the outcomes by [58].

While certain earlier researchers evaluated the restricted realism of datasets, this review recognises their foundational role in developing best practices and baseline performance benchmarks. In addition, it explores how public datasets contribute to consistency by encouraging transparency and replication. These outcomes recommend that continued investment in high-quality and diverse datasets is important for advancing churn modelling studies and enables practical advancements in customer retention analytics [21, 35].

From the research, the gaps are summarised as,

- Limited comparative studies of XAI techniques: Most studies focus on individual methods like SHAP or LIME

without comparing their interpretability, fidelity, or stakeholder usability. There is a lack of benchmarking frameworks to evaluate XAI methods across industries (e.g. telecom vs. banking).

- **Underexplored profit-based evaluation:** Churn models often optimise for accuracy or AUC, ignoring profitability metrics like Customer Lifetime Value (CLV) or retention ROI.
- **Sparse integration of human-centred design:** Few models incorporate stakeholder feedback loops or usability testing for explanations. Limited exploration of how different user groups (e.g. analysts vs. customer service reps) interpret XAI outputs.
- **Bias detection and fairness:** XAI is rarely used to uncover or mitigate bias in churn prediction models, despite regulatory importance.
- **Real-time and prescriptive capabilities:** Most XAI applications are post hoc and static; real-time explainability and prescriptive optimisation (e.g. budget allocation for retention) are underdeveloped.
- **Lack of domain-specific datasets:** Many studies rely on synthetic or generic datasets, limiting applicability to specific sectors like digital banking or telecom.

The recommendations are:

- **Develop cross-industry XAI evaluation frameworks:** The research creates standardised metrics for interpretability, stability and stakeholder usability. This review encourages comparative studies of SHAP, LIME and newer methods like integrated gradients.
- **Incorporate profit-based metrics:** The review suggested using CLV and churn cost in model evaluation to align predictions with business impact.
- **Design human-centred XAI interfaces:** Build dashboards that allow non-technical users to interact with explanations. There is a need to include feedback mechanisms to refine model behaviour based on user input.
- **Use XAI for bias auditing:** Apply feature attribution methods to detect discriminatory patterns. There is a need for an integration of fairness constraints during model training and explanation generation.
- **Enable real-time and prescriptive AI:** There is a need for a combination of XAI with optimisation techniques to recommend retention actions. There is a suggestion to use fuzzy clustering and LLMs to generate contextual retention strategies.
- **Promote open, domain-specific datasets:** This review encourages data sharing across industries to improve model generalisability. This review develops benchmarks tailored to churn prediction in sectors like finance, telecom and e-commerce.

The managerial implications are,

Explainable AI (XAI) techniques provide actionable insights that help to design retention strategies tailored to industry-specific contexts:

- **Banking and insurance:** Interpretable models such as GAMs and decision trees support compliance and audit requirements. SHAP explanations document risk factors transparently, enabling managers to align retention policies with regulatory standards while maintaining customer trust.
- **Telecommunications:** SHAP and counterfactual explanations highlight key churn drivers (e.g. pricing, service quality). Managers can use these insights to craft personalised retention offers and monitor attribution drift to ensure strategies remain effective under market changes.
- **Retail and E-commerce:** Global SHAP narratives and LIME explanations clarify why customers belong to specific segments, guiding targeted promotions. Counterfactuals help managers design feasible next-best-actions (e.g. discounts, loyalty rewards) that balance retention with profitability.

Across industries, XAI enhances retention management by making customer drivers transparent, ensuring strategies are both effective and auditable, and translating complex models into actionable narratives for decision makers [7, 26, 53].

5 Conclusion

This systematic literature review emphasised the transformative potential of XAI in customer analytics, specifically in churn prediction, segmentation and retention. By bridging the gap between interpretability and predictive performance, XAI empowers sectors to make transparent, informed and ethically better decisions. The analysis of previous research explores a growing trend towards incorporating interpretability tools into business intelligence systems, with SHAP and LIME emerging as dominant techniques in addition to several ensemble models. But the challenges remain especially in the areas of real-time deployment, lack of standardised interpretability metrics and cross-domain generalisability. Future research must focus on developing domain-adapted and scalable XAI contexts and explore the intersection of explainability with fairness, user experience and privacy. Eventually, the XAI adoption in customer analytics is not said to be a technical upgrade, and it is a strategic imperative for effective and responsible data-driven decision-making.

Author contribution M.M. (Misbah Maqbool) conceptualized the study, designed the methodology, conducted the systematic literature

review, and wrote the main manuscript text. I.H.Q. (Ishtiaq Hus-sain Qureshi) contributed to data collection, analysis, interpretation of results, and reviewed and edited the manuscript. All authors have read and approved the final version of the manuscript.

Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interests The authors declare no conflict of interests.

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